

The Pais–Uhlenbeck Metric as a Minimum-Distortion Principle, and the Representation Problem for the Fourth-Order Field

Abstract

For the unequal-frequency PT-symmetric Pais–Uhlenbeck oscillator, the admissible complex-symplectic diagonalizers form a coset $S_+ SO(2, \mathbb{C})^2$ of the unique Hermitian positive-definite diagonalizer S_+ . We prove that S_+ minimizes the affine-invariant metric distortion: for every admissible S , $\|\log(S^\dagger S)\|_F^2 \geq 4r^2$ with $r = \log \frac{\omega_1 + \omega_2}{\omega_1 - \omega_2}$, with equality exactly on the unitary stabilizer coset, on which the induced Hilbert-space metric is constantly the Bender–Mannheim $\eta = e^{-Q}$. The proof is elementary and exact: at worst-case stabilizer alignment the log-eigenvalues of $S^\dagger S$ are $\operatorname{arccosh}(\cosh r \cosh b) \pm a$ in the Gram rapidities (a, b) of the stabilizer factor. We then quantize the free fourth-order field modewise and compute the Bender–Mannheim ground state exactly: it is a normalizable real Gaussian for every mode, but its fidelity with the two-field Fock vacuum tends to the universal constant $\sqrt{3}/2$ at large momentum, and its relative particle content to $1/3$ per mode pair. Consequently the physical (PT) and naive Fock representations of the free fourth-order field are disjoint in every spatial dimension, although each mode is unitarily equivalent: PT unitarization is exact modewise and changes the field-theoretic representation. All identities are machine-verified.

1 Introduction

This paper continues the symplectic analysis of the PT-symmetric Pais–Uhlenbeck (PU) oscillator begun in [1], whose results and conventions we use freely: $V = (x, y, p, q)^\top$, $[V_a, V_b] = iJ_{ab}$, $H_{\text{PT}} = \frac{1}{2}V^\top GV$ with the PU matrix G , positive normal form G_0 , rapidity

$$r = \log \frac{\omega_1 + \omega_2}{\omega_1 - \omega_2}, \quad \omega_1 > \omega_2 > 0, \quad (1)$$

and, after the canonical normalization $d_x d_y = \gamma \omega_1 \omega_2$, the unique Hermitian positive-definite diagonalizer

$$S_+ = \cosh \frac{r}{2} I + \sinh \frac{r}{2} B \in \operatorname{Sp}(4, \mathbb{C}), \quad S_+^\top G' S_+ = G'_0, \quad S_+^\top J S_+ = J, \quad (2)$$

with B a fixed Hermitian involution. The full solution family of the diagonalization problem is the coset $S_+ \operatorname{Stab}(J, G'_0)$ with $\operatorname{Stab}(J, G'_0) \cong SO(2, \mathbb{C}) \times SO(2, \mathbb{C})$, and the metaplectic logarithm of S_+ is the Bender–Mannheim generator $Q = \alpha p q + \beta x y$ [2, 3, 1].

Two questions are addressed here.

First, uniqueness of the positive diagonalizer does not by itself distinguish S_+ variationally: one may ask whether S_+ *minimizes* a natural functional over all admissible diagonalizers. We prove that it minimizes the affine-invariant distortion $\mathcal{F}(S) = d(I, S^\dagger S)^2 = \|\log(S^\dagger S)\|_F^2$ (Theorem 2.5), so that the Bender–Mannheim metric is selected by a minimum-distortion principle:

it is the least-distorting complex canonical transformation converting the PU Hamiltonian to its positive normal form.

Second, for the free fourth-order field $(\square + m_1^2)(\square + m_2^2)\phi = 0$ each spatial momentum k contributes a PU pair with $\omega_i(k) = \sqrt{k^2 + m_i^2}$, and the modewise construction applies with rapidity

$$r(k) = \log \frac{(\omega_1(k) + \omega_2(k))^2}{m_1^2 - m_2^2} = 2 \log |k| + O(1) \quad (m_1 > m_2). \quad (3)$$

One might expect from (3) that the modewise transformation carries unboundedly growing squeezing and fails Fock implementability “dramatically”. The exact computation shows something sharper and more structured (Theorems 3.2–3.4): the PT ground state exists as a normalizable Gaussian for *every* mode, its mismatch with the free two-field vacuum saturates at *universal constants* — fidelity $\sqrt{3}/2$ and relative occupation $1/3$ per mode pair at large k — and therefore the physical and naive representations are disjoint in every spatial dimension $d \geq 1$, with vacuum overlap decaying like $e^{-cV\Lambda^d}$ under a cutoff. The divergence of $r(k)$ itself is a statement about the observable-space (phase-space) metric, which we separate cleanly (Corollary 3.1): its eigenvalues $e^{\pm r(k)} \asymp (1 + k^2)^{\pm 1}$ make the phase-space metric an $H^1 \oplus H^{-1}$ -type Sobolev pair, unboundedly inequivalent to the standard norm, while the state-level mismatch remains $O(1)$ per mode.

Verification. As in [1], every finite-dimensional identity in this paper — the invariant formulas, the closed-form log-eigenvalues, the PT ground-state Gaussian, the eigenvalue equation, the fidelity and occupation formulas, and the limits $\sqrt{3}/2$ and $1/3$ — is machine-verified symbolically, with independent numerical checks (global optimization for Theorem 2.5; high-precision quadrature for Theorem 3.2). The scripts accompany the verification repository of [1].

2 The minimum-distortion theorem

Work in the canonically normalized coordinates of [1]; all adjoints refer to them. The stabilizer is parametrized by two complex angles,

$$C(\theta_1, \theta_2) = \sum_{j=1,2} (\cos \theta_j P_j + \sin \theta_j X_j), \quad X_j^2 = -P_j, \quad (4)$$

with P_j the projections onto the modes of G'_0 and X_j the per-mode complex structures; every admissible diagonalizer is $S = S_+ C$.

Definition 2.1. For an admissible S let $\mathcal{F}(S) = \|\log(S^\dagger S)\|_F^2$, the squared affine-invariant distance of the observable-space metric $S^\dagger S$ from the identity [4]. For a stabilizer element $C = C_1 \oplus C_2$ (blocks on the two modes) define its *Gram rapidities* $\tau_j \geq 0$ by

$$W_j = C_j C_j^\dagger, \quad \det W_j = 1, \quad \cosh \tau_j = \frac{1}{2} \operatorname{tr} W_j, \quad (5)$$

and set $a = \frac{1}{2}(\tau_1 + \tau_2)$, $b = \frac{1}{2}(\tau_1 - \tau_2)$.

Note $\tau_1 = \tau_2 = 0$ iff $W_1 = W_2 = I$ iff C is unitary. Since $P = S^\dagger S = C^\dagger S_+^2 C$ is Hermitian positive *and* symplectic, its eigenvalues form reciprocal pairs $e^{\pm x_1}, e^{\pm x_2}$ with $x_i \geq 0$, and $\mathcal{F}(S) = 2(x_1^2 + x_2^2)$.

Lemma 2.2 (Invariants). *Write $W = CC^\dagger = W_1 \oplus W_2$, $c = \cosh r$, $s = \sinh r$. Decomposing $W_j = \cosh \tau_j I + \sinh \tau_j (n_j \cdot \sigma)$ with real unit vectors n_j , and letting $\chi \in [-1, 1]$ denote the alignment $n_1 \cdot \tilde{n}_2$ (where $\tilde{n}_2$ is n_2 with its σ_2 component reflected),*

$$\cosh x_1 + \cosh x_2 = c (\cosh \tau_1 + \cosh \tau_2), \quad (6)$$

$$\cosh x_1 \cosh x_2 = \frac{1}{2} \left[(2c^2 - s^2) \cosh \tau_1 \cosh \tau_2 + s^2 (1 + \sinh \tau_1 \sinh \tau_2 \chi) \right]. \quad (7)$$

Proof. $\text{tr } P = \text{tr}(e^r B W) = c \text{tr } W + s \text{tr}(BW)$, and $\text{tr}(BW) = 0$ because B is off-diagonal and W diagonal with respect to the mode splitting; this gives (6). For (7), expand $\text{tr } P^2 = c^2 \text{tr } W^2 + s^2 \text{tr}((BW)^2)$ (the cross terms again vanish blockwise); the off-diagonal blocks of B equal $-\sigma_2$, and the 2×2 identity $\sigma_2 M \sigma_2 = (\det M) M^{-\top}$ reduces $\text{tr}((BW)^2) = 2 \text{tr}(\overline{W_2}^{-1} W_1) = 4 [\cosh \tau_1 \cosh \tau_2 - \sinh \tau_1 \sinh \tau_2 \chi]$. Converting $(\text{tr } P, \text{tr } P^2)$ to the symmetric functions of $\cosh x_i$ yields (7). (Both matrix identities are verified against random stabilizer elements in the repository.) $\square$

Lemma 2.3 (Worst alignment; closed form). *At $\chi = -1$ the system (6)–(7) is solved exactly by*

$$x_{1,2} = \varphi(b) \pm a, \quad \varphi(b) = \text{arccosh}(\cosh r \cosh b) \geq r. \quad (8)$$

Proof. With $\alpha = \cosh a$, $\beta = \cosh b$ one finds from (6)–(7) at $\chi = -1$, using $\cosh \tau_1 \cosh \tau_2 \pm \sinh \tau_1 \sinh \tau_2 = \cosh(\tau_1 \pm \tau_2)$,

$$\cosh x_1 + \cosh x_2 = 2c\alpha\beta, \quad \cosh x_1 \cosh x_2 = \alpha^2 + c^2\beta^2 - 1,$$

whence $\cosh x_{1,2} = c\alpha\beta \pm \sqrt{(\alpha^2 - 1)(c^2\beta^2 - 1)} = \cosh(\varphi \pm a)$ with $\cosh \varphi = c\beta$. Both hyperbolic identities are verified symbolically. Finally $\cosh \varphi(b) - \cosh r = \cosh r(\cosh b - 1) \geq 0$ gives $\varphi(b) \geq r$ with equality iff $b = 0$. $\square$

Lemma 2.4 (Alignment monotonicity). *At fixed $\cosh x_1 + \cosh x_2$, the quantity $x_1^2 + x_2^2$ is nondecreasing in $\cosh x_1 \cosh x_2$; and (7) is nondecreasing in χ .*

Proof. Parametrize $\cosh x_{1,2} = T/2 \pm w$, $w \geq 0$; the product decreases in w , and $\frac{d}{dw}(x_1^2 + x_2^2) = 2\left(\frac{x_1}{\sinh x_1} - \frac{x_2}{\sinh x_2}\right) \leq 0$ because $x/\sinh x$ is strictly decreasing and $x_1 \geq x_2$. The second statement is the sign of the coefficient $s^2 \sinh \tau_1 \sinh \tau_2 / 2 \geq 0$ in (7). $\square$

Theorem 2.5 (Minimum distortion). *For every admissible diagonalizer $S = S_+ C$,*

$$\mathcal{F}(S) \geq 4 \left[a^2 + \text{arccosh}^2(\cosh r \cosh b) \right] \geq 4r^2 = \mathcal{F}(S_+), \quad (9)$$

with overall equality if and only if C is unitary. All minimizers share the polar factor S_+ , hence the same Hilbert-space metric $\eta = e^{-Q}$: the Bender–Mannheim metric is the unique metric of any minimum-distortion diagonalizer,

$$S_+ \in \arg \min_{S^\top J S = J, S^\top G' S = G'_0} d(I, S^\dagger S), \quad Q = -2 \log(\text{polar factor of any minimizer})|_{\text{metaplectic}}. \quad (10)$$

Proof. By Lemma 2.4, lowering χ to -1 at fixed Gram rapidities does not increase $x_1^2 + x_2^2$; by Lemma 2.3 the value there is $\mathcal{F} = 2[(\varphi + a)^2 + (\varphi - a)^2] = 4(\varphi^2 + a^2)$; and $\varphi(b) \geq r$. Equality in the chain forces $a = 0$ and $b = 0$, i.e. $\tau_1 = \tau_2 = 0$, i.e. C unitary; the converse is immediate since then $S^\dagger S$ is unitarily conjugate to S_+^2 . If C is unitary, $S_+ C = S_+ \cdot C$ is already a polar decomposition, so the polar factor is S_+ and the metaplectic implementer factorizes as $\rho' = U_C e^{-Q/2}$ with U_C unitary, giving $\eta' = \rho'^\dagger \rho' = e^{-Q}$ for every minimizer. $\square$

Remark 2.6. (i) The individual log-eigenvalues do *not* satisfy $x_i \geq r$: numerical scans show $\min_i x_i < r$ on open regions of the stabilizer; only the joint quantity is bounded below, which is why the two-invariant argument is needed. (ii) The equality set is the unitary part of the stabilizer, which in the canonical coordinates is generically the finite group $\mathbb{Z}_2 \times \mathbb{Z}_2$ [1]; abstractly it is contained in the maximal compact $U(1)^2$ of $SO(2, \mathbb{C})^2$. (iii) The theorem gives the finite-dimensional Kempf–Ness-type picture concretely: the Gram map $C \mapsto CC^\dagger$ measures the distance of a coset element from the unitary locus, and (8) shows how the two rapidities (a, b) add in quadrature with the PU rapidity r .

3 The free fourth-order field

3.1 Modewise data and the phase-space Hilbert scale

For $(\square + m_1^2)(\square + m_2^2)\phi = 0$, each spatial momentum k carries a PU pair with $\omega_i(k) = \sqrt{k^2 + m_i^2}$ and $m_1 > m_2 > 0$. Since $(\omega_1 - \omega_2)(\omega_1 + \omega_2) = m_1^2 - m_2^2$ identically, the rapidity obeys (3) *exactly*, interpolating between $r(0) = \log \frac{m_1 + m_2}{m_1 - m_2}$ and $r(k) = 2 \log |k| + O(1)$.

Corollary 3.1 (Phase-space Hilbert scale). *In the canonically normalized modewise coordinates, the observable-space metric $M_{\text{obs}}(k) = S_+(k)^\dagger S_+(k)$ has eigenvalues $e^{\pm r(k)}$ with*

$$e^{r(k)} = \frac{(\omega_1(k) + \omega_2(k))^2}{m_1^2 - m_2^2} \asymp 1 + k^2, \quad e^{-r(k)} \asymp (1 + k^2)^{-1}. \quad (11)$$

On the spectral subspaces $\Pi_\pm(k)$ of B the induced one-particle phase-space norm is therefore equivalent to an $H^1 \oplus H^{-1}$ Sobolev pair (up to the constant $m_1^2 - m_2^2$ and bounded factors): it is not uniformly equivalent to the standard norm, and no scalar renormalization makes it so.

This is the precise content of the observation that the modewise metric “blows up in the UV”. It concerns the classical (phase-space) metric. The quantum state-level comparison is finer, and is computed next.

3.2 The PT ground state, exactly

Fix a mode and work in the normalized coordinates (x, y) with $\omega_1 > \omega_2 > 0$. The positive Hamiltonian $\tilde{H}'$ has ground state $\phi_0 \propto e^{-\frac{1}{2}(x^2/\omega_2 + \omega_1 y^2)}$. The PT Hamiltonian in these coordinates is the differential operator

$$H'_{\text{PT}} = \frac{1}{2} \left[\frac{\omega_1^2 + \omega_2^2}{\omega_1 \omega_2} x^2 + \omega_1 \omega_2 y^2 - \omega_1 \omega_2 \partial_x^2 \right] - x \partial_y. \quad (12)$$

Theorem 3.2 (PT ground state). *The unique Gaussian solution of the modewise annihilation equations $(\ell_j^\top S_+^{-1} V)\psi_0 = 0$ is*

$$\psi_0(x, y) \propto \exp\left[-\frac{1}{2}\left(\frac{\omega_1 + \omega_2}{\omega_1 \omega_2} x^2 + 2xy + (\omega_1 + \omega_2) y^2\right)\right], \quad (13)$$

a real Gaussian with $\det(\text{quadratic form}) = \frac{\omega_1^2 + \omega_1 \omega_2 + \omega_2^2}{\omega_1 \omega_2} > 0$: normalizable for all $\omega_1 > \omega_2 > 0$, and $H'_{\text{PT}}\psi_0 = \frac{\omega_1 + \omega_2}{2}\psi_0$. Its fidelity with the two-oscillator vacuum and its occupation relative to it are exactly

$$f(\omega_1, \omega_2) = |\langle \hat{\phi}_0 | \hat{\psi}_0 \rangle|^2 = \frac{4\omega_1 \sqrt{\omega_1^2 + \omega_1 \omega_2 + \omega_2^2}}{4\omega_1^2 + 3\omega_1 \omega_2 + \omega_2^2}, \quad (14)$$

$$\langle N \rangle(\omega_1, \omega_2) = \langle \hat{\psi}_0 | (N_1 + N_2) | \hat{\psi}_0 \rangle = \frac{\omega_2(\omega_1^2 + \omega_2^2)}{2\omega_1(\omega_1^2 + \omega_1 \omega_2 + \omega_2^2)}, \quad \langle N_1 \rangle = \langle N_2 \rangle. \quad (15)$$

In particular, along the field dispersion $\omega_i = \omega_i(k)$,

$$\lim_{|k| \rightarrow \infty} f = \frac{\sqrt{3}}{2} \approx 0.866, \quad \lim_{|k| \rightarrow \infty} \langle N \rangle = \frac{1}{3}, \quad (16)$$

universal constants independent of m_1, m_2, γ .

Proof. Solving the two first-order equations for a Gaussian ansatz gives the stated quadratic form (one solution); the eigenvalue equation is a direct substitution. The fidelity is the standard two-Gaussian determinant formula; the occupations follow from covariance algebra, $\langle N_j \rangle = \frac{1}{2} c_j^\top \Sigma c_j$ with Σ the covariance of $|\psi_0|^2$ and c_j the coefficient vectors of $a_j \psi_0$. All steps are verified symbolically, and the fidelity is confirmed by 25-digit quadrature. The limits follow with $\omega_1/\omega_2 \rightarrow 1$. $\square$

Remark 3.3. Two features deserve emphasis. (i) ψ_0 is a smooth, normalizable Gaussian even as $\omega_1 \rightarrow \omega_2$: the exceptional-point pathology (divergence of S_+ , Jordan structure) lives in the similarity transformation and the excited spectrum, not in the ground state itself. (ii) A naive Bogoliubov estimate would assign the modewise map a squeezing $\sim \sinh^2(r(k)/2) \sim k^2$ and hence a violently divergent particle content per mode. The exact result (16) is $O(1)$: the diverging rapidity $r(k)$ is carried almost entirely by directions that do not create quanta. The naive estimate fails because $\rho = e^{-Q/2}$ is not $*$ -preserving on the original Fock structure — the transformed “annihilators” $\rho a_j \rho^{-1}$ do not satisfy the unitary Bogoliubov normalization ($\alpha \alpha^\dagger - \beta \beta^\dagger = \cosh r \neq 1$) — so Shale–Stinespring-type criteria cannot be applied to ρ directly. Comparing the two *states*, both of which live in the same L^2 , avoids this trap entirely.

3.3 Disjointness of the representations

Theorem 3.4 (Representation change). *Let $m_1 > m_2 > 0$, $d \geq 1$, and consider the free fourth-order field in a box of volume V with momentum cutoff Λ , with $\Psi_{\text{PT}} = \bigotimes_k \hat{\psi}_0(k)$ the modewise PT ground state and $\Omega = \bigotimes_k \hat{\phi}_0(k)$ the free two-field Fock vacuum. Then:*

1. *the vacuum overlap satisfies $|\langle \Omega | \Psi_{\text{PT}} \rangle|^2 = \prod_k f(k) \leq e^{-cV\Lambda^d}$ for some $c > 0$ depending only on m_1/m_2 and d (since $f(k) \rightarrow \sqrt{3}/2 < 1$);*

2. the relative particle density diverges in the UV: $\frac{1}{V}\langle N_{\text{tot}} \rangle = \int^\Lambda \frac{d^d k}{(2\pi)^d} \langle N \rangle(k) \sim \frac{1}{3} \int^\Lambda \frac{d^d k}{(2\pi)^d} = \infty$ as $\Lambda \rightarrow \infty$, in every dimension $d \geq 1$;
3. in the thermodynamic/continuum limit the incomplete-tensor-product classes of Ψ_{PT} and Ω differ ($\sum_k (1 - |\langle \hat{\psi}_0 | \hat{\phi}_0 \rangle|) = \infty$), so the GNS representations of the CCR algebra built on the two states are disjoint: no density matrix in the Fock representation of the free pair (m_1, m_2) reproduces the PT vacuum, and vice versa.

Each mode, by contrast, is unitarily equivalent: the modewise Dyson maps $\rho_k = e^{-Q_k/2}$ define unitaries between the mode Hilbert spaces equipped with the physical (η_k) and standard inner products.

Proof. (1) and (2) are immediate from Theorem 3.2: $f(k)$ and $\langle N \rangle(k)$ are continuous in k with the universal limits (16), so beyond some $k_*(m_1/m_2)$ one has, e.g., $f(k) \leq 0.9$ and $\langle N \rangle(k) \geq 0.3$. (3) is von Neumann’s equivalence criterion for incomplete tensor products [5] applied to the two product vectors, combined with purity: inequivalent pure product states generate disjoint representations. The modewise statement is the finite-dimensional similarity of [3, 1]. $\square$

Remark 3.5 (What this does and does not say). The theorem does not say the PT quantization of the fourth-order field is inconsistent. The physical Hilbert space, defined as the completion of the modewise-unitarized theory, is naturally identified with the Fock space of the free pair (m_1, m_2) ; on it the dynamics is unitary, exactly as argued by Bender and Mannheim modewise. What fails is the *global* equivalence between this physical representation and the naive covariant Fock representation in which the fourth-order propagator and perturbation theory are usually written: the two are different, mutually singular representations of the same field algebra, with an $O(1)$ discrepancy per UV mode — not a discrepancy that decays with k , and not one confined to the exceptional locus $m_1 = m_2$. Any interacting construction therefore has to choose a representation first; formal manipulations that mix them (e.g., inserting $\eta = e^{-Q}$ as if it were a bounded operator on the covariant Fock space) are not supported: by Corollary 3.1 the metric is unbounded with unbounded inverse, defining a Hilbert scale rather than an equivalent norm, and by Theorem 3.4 its formal infinite-mode product annihilates the covariant vacuum’s representation class entirely. This sharpens, for the free theory, the sense in which PT-based unitarity claims for quadratic gravity [6] require an explicit domain-and-completion construction.

4 Discussion

The variational principle. Theorem 2.5 upgrades the uniqueness theorem of [1]: the Bender–Mannheim metric is not merely the unique positive point of the diagonalizer coset but its unique distortion minimizer, with the exact excess $4[a^2 + \text{arccosh}^2(\cosh r \cosh b)] - 4r^2$ in the Gram rapidities of the stabilizer. The result is a concrete, fully explicit instance of the Kempf–Ness/Mostow picture — minimal-norm points on complexified orbits are unitary-locus points — but the proof is elementary and yields the minimizing value in closed form, which the abstract theory does not.

The two geometries, again. The field-theoretic analysis separates two statements that are easily conflated. The *phase-space* metric degenerates in the UV exactly as the rapidity formula (3) dictates ($e^{\pm r(k)} \asymp k^{\pm 2}$, a Sobolev pair $H^1 \oplus H^{-1}$); this is a statement about observables. The *state-space* mismatch saturates at universal constants ($\sqrt{3}/2$, $1/3$); this is a statement about representations, and it is what controls implementability. Both are exact consequences of the same modewise data. The constants are universal because the UV limit $\omega_1/\omega_2 \rightarrow 1$ erases all mass and coupling dependence — remarkably, the same limit in which the finite-dimensional theory degenerates (the exceptional point) is the limit in which the ground-state data become featureless.

Outlook. Three natural continuations: (i) extend the minimum-distortion theorem to general semisimple quadratic PT Hamiltonians, where the stabilizer is a product of $SO(2, \mathbb{C})$ factors and the same Gram-rapidity reduction should apply mode-pair by mode-pair; (ii) determine whether the disjointness survives interactions in low dimensions, where the free-theory obstruction is only the universal $1/3$ per mode (a wave-function renormalization of the modewise map, $\rho_k \rightarrow \rho_k Z_k$ with Z_k unitary, cannot change class (3), but a k -dependent redefinition of the mode split might); and (iii) make contact with the Krein-space formulation of indefinite-metric QFT, where the covariant representation lives, to phrase the representation choice intrinsically. We leave these to future work.

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